

Team Aqua

Analyzing Olympic Data

1996-2016

Agenda

01 Overview of Data

02 Data Cleanup

03 Exploratory Data Analysis

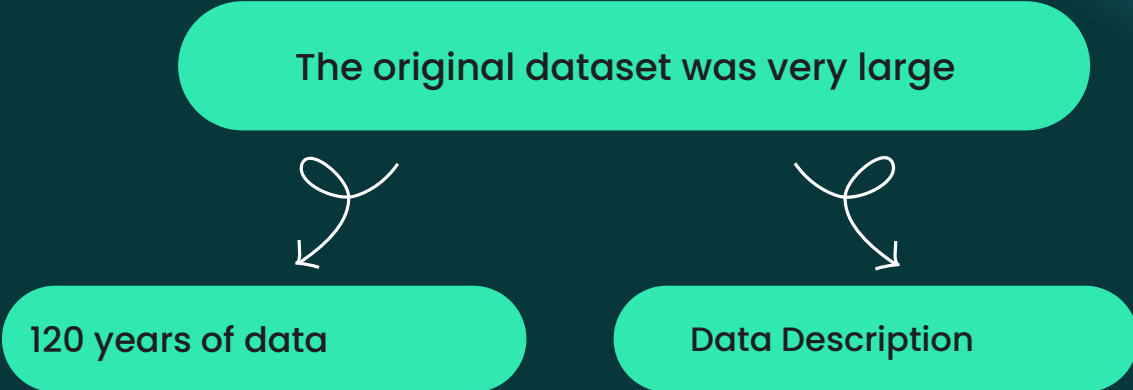
04 Data Aggregations

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06 Conclusion

FULL dataset overview

The original dataset was very large



```
graph TD; A([The original dataset was very large]) --> B([120 years of data]); A --> C([Data Description]);
```

120 years of data

- Over 270,000+ rows
- Included countries that no longer exist and have been renamed

Data Description

- Contained 18 columns
- 2 spreadsheets
- Basic bio data on athletes and medal results from Athens 1896 to Rio 2016

Data Cleaning Steps

Step 1: Join Data

We left joined on the National Olympic Committee column, it is the common key that uniquely identifies each country or team in both datasets

Step 2: Filter Data

For readability, we filtered the years from 1996-2016 to focus on recent data.

Step 3: Metric Conversions

Converted the height and weight from kg to lbs, and cm to inches for better readability

Step 4: Filled Null Values

We handled missing values by replacing categorical NaNs with meaningful labels like 'No Medal' and 'Unknown'

Step 5: Validated output

We verified data completeness by checking for remaining missing values using `.isna().sum()`

SUMMARY STATISTICS

- Overview of the summary of our filtered data, with new columns, and datatype
- Shows the count of rows we are analyzing went from nearly 270k -> 101k

#	Column	Non-Null Count	Dtype
0	ID	101,163	int64
1	Name	101,163	object
2	Sex	101,163	object
3	Age	101,163	float64
4	Height_cm	101,163	float64
5	Weight_kg	101,163	float64
6	Team	101,163	object
7	NOC	101,163	object
8	Games	101,163	object
9	Year	101,163	int64
10	Season	101,163	object
11	City	101,163	object
12	Sport	101,163	object
13	Event	101,163	object
14	Medal	101,163	object
15	region	101,163	object
16	Weight_lbs	101,163	float64
17	Height_in	101,163	float64
18	Has_Medal	101,163	int64

ANALYSIS QUESTIONS

Who is competing in the Olympics?

Who is winning?

**How does this change across seasons,
countries, and genders?**

DATA AGGREGATION

When aggregating our data, we were looking at the following questions:

- **How many total athletes competed in the Olympic Games from 1996-2016?**
- **How many different sporting events were there to compete in?**

101,163
athlete entries

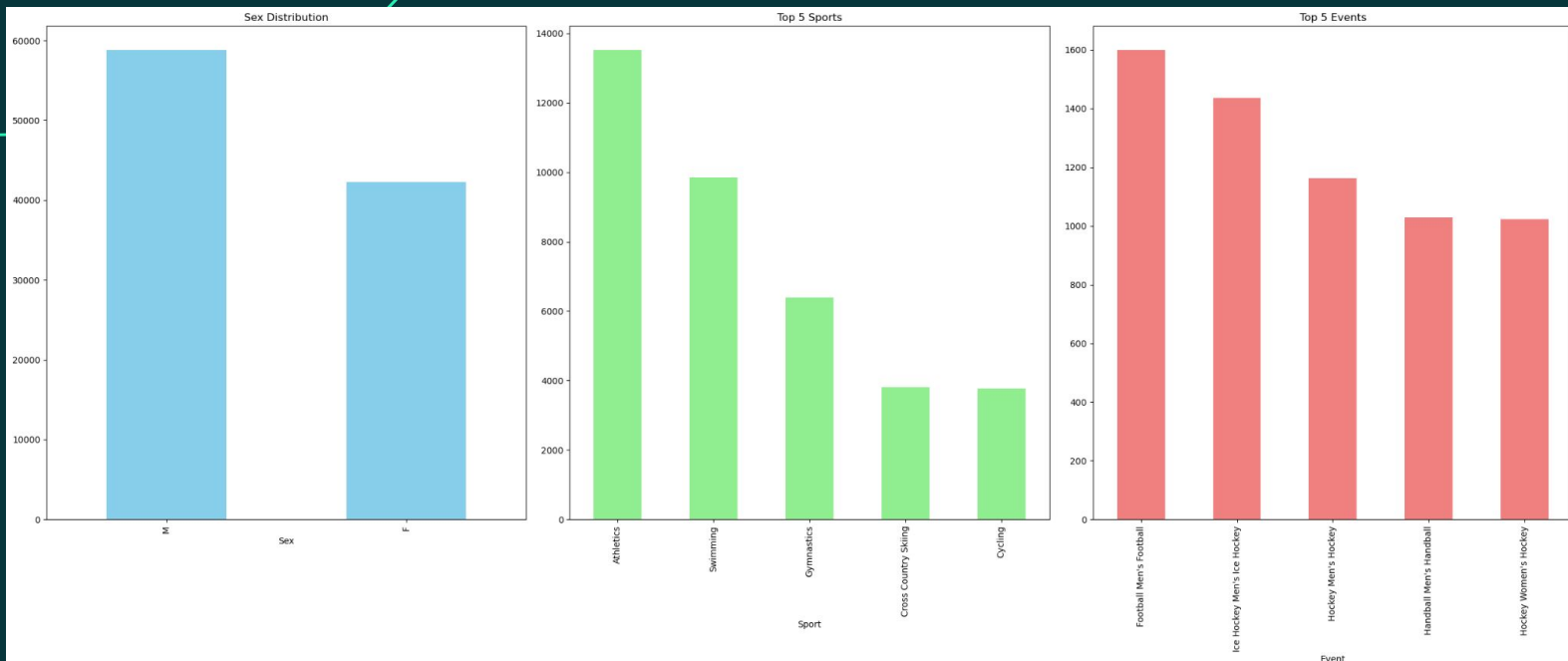
52,071
unique athletes

51
sports

243
competing teams
(Some teams are not nations)

DISTRIBUTION CHARTS

Men outnumber women across the dataset, though the gap has been narrowing as more women's events are introduced. Athletics, swimming and gymnastics drive the highest entry counts due to the sheer number of individual events they offer. Among specific events team spots like Men's Football and Men's Ice Hockey lead.



DISTRIBUTION HISTOGRAMS

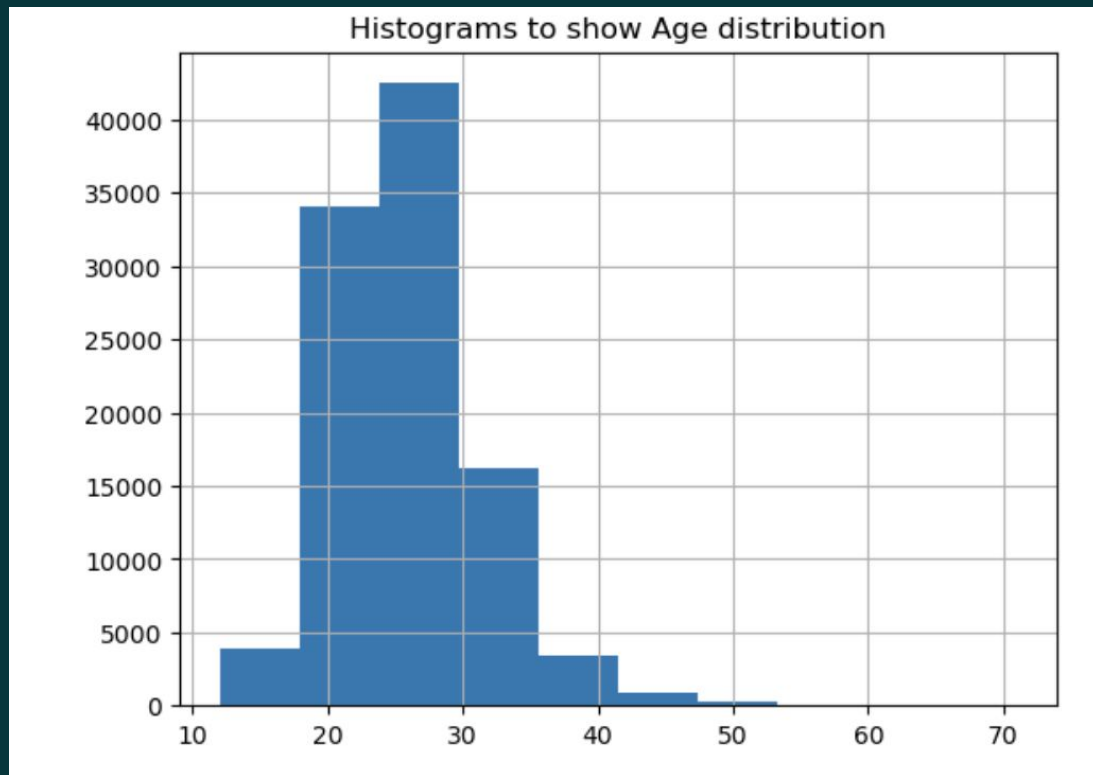
The age distribution is right skewed, with most athletes between 20 and 30 years old. A small tail extends past 50, these are typically participants in precision sports like equestrianism and shooting.

25.7 yrs

Mean athlete age

12 yrs

Youngest
competitor



DATA AGGREGATION

Medal distribution shifts significantly between seasons. In summer the US leads with 1,557 medals, nearly double second place Russia. In Winter, Canada takes the top spot while countries like Finland emerge as Winter powers, despite a minimal summer presence. Geography, climate and national investment shape medal outcomes.

Summer

Medal	Gold	Silver	Bronze	Total
Team				
United States	799	397	361	1557
Russia	292	265	313	870
Australia	213	274	328	815
Germany	244	185	300	729
China	280	242	168	690

Winter

Medal	Gold	Silver	Bronze	Total
Team				
Canada	222	100	46	368
United States	67	166	98	331
Germany	96	95	52	243
Russia	75	76	73	224
Finland	11	54	127	192

DATA AGGREGATION

Summer Sports by Sex

Overall, more men have competed in the Olympics than women have, but some sports are played about equally by men and women, while others are more skewed. Softball, Synchronized Swimming, Rhythmic Gymnastics, and Baseball are single-gender sports at the Summer Olympics.

Close to Even

Sex	F	M	Total	M:F Ratio
Sport				
Softball	478	0	478	0.00
Synchronized Swimming	658	0	658	0.00
Rhythmic Gymnastics	544	0	544	0.00
Basketball	853	707	1560	0.83
Badminton	630	601	1231	0.95

Mostly Men

Sex	F	M	Total	M:F Ratio
Sport				
Cycling	1198	2578	3776	2.15
Canoeing	807	1939	2746	2.40
Wrestling	304	1562	1866	5.14
Boxing	72	1710	1782	23.75
Baseball	0	731	731	inf

DATA AGGREGATION

Winter Sports by Sex

More men than women have competed in every Winter sport, although some are close to an even split. The Nordic Combined is the only single-sex sport at the Winter Games.

Close to Even

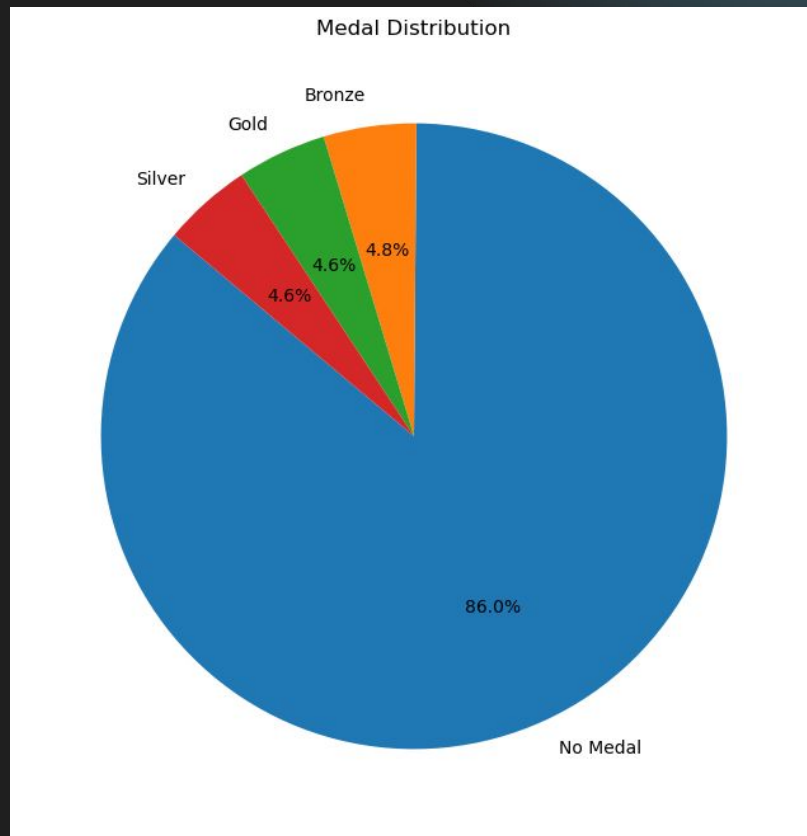
Sex	F	M	Total	M:F Ratio
Sport				
Figure Skating	400	402	802	1.00
Curling	222	225	447	1.01
Short Track Speed Skating	609	613	1222	1.01
Biathlon	1472	1572	3044	1.07
Speed Skating	853	927	1780	1.09

Mostly Men

Sex	F	M	Total	M:F Ratio
Sport				
Ice Hockey	754	1436	2190	1.90
Luge	160	425	585	2.66
Bobsleigh	143	892	1035	6.24
Ski Jumping	30	896	926	29.87
Nordic Combined	0	614	614	inf

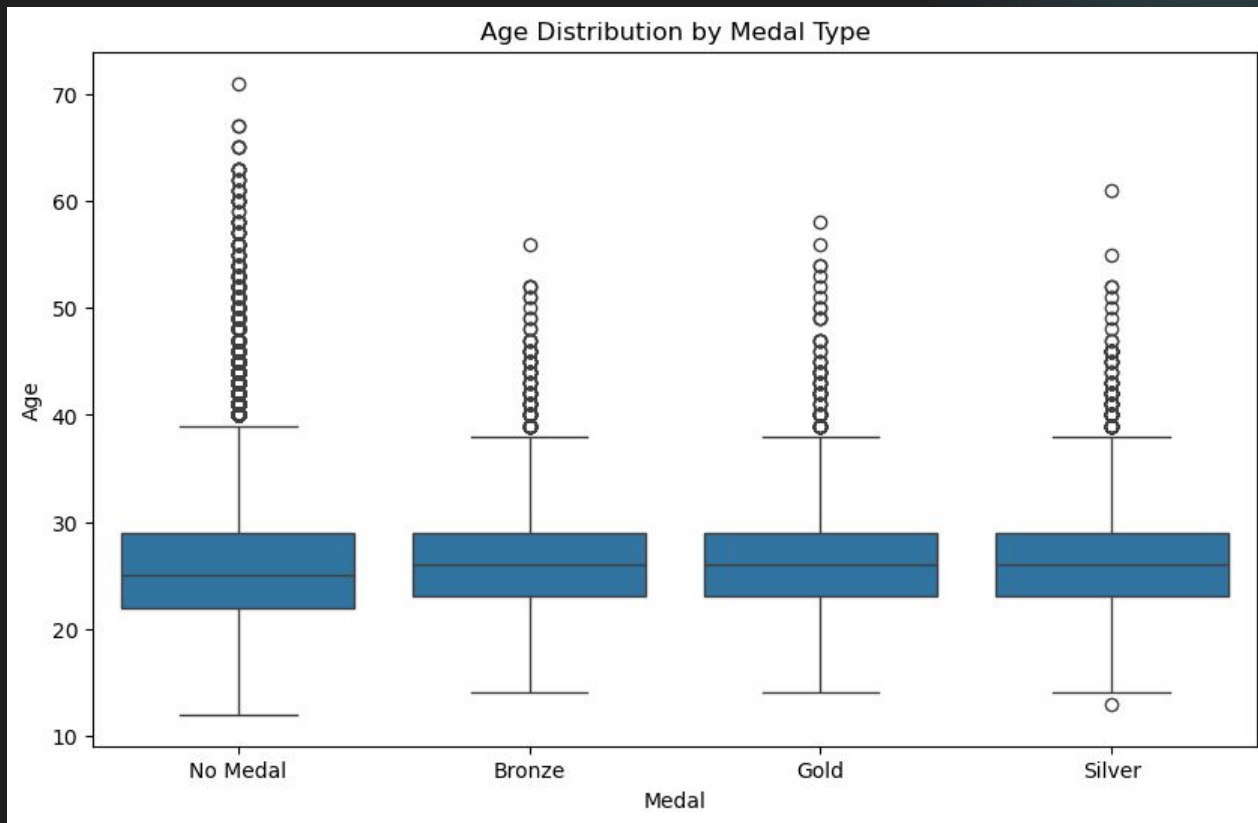
Data Visualization: Medal Distribution

For any given event, only about 14% of competitors will win a medal of any kind. The vast majority of Olympic athletes go home empty handed. Gold, silver and Bronze are distributed nearly equally, with Bronze slightly ahead. That edge probably represents events where ties or third place matches awarded multiple bronzes.



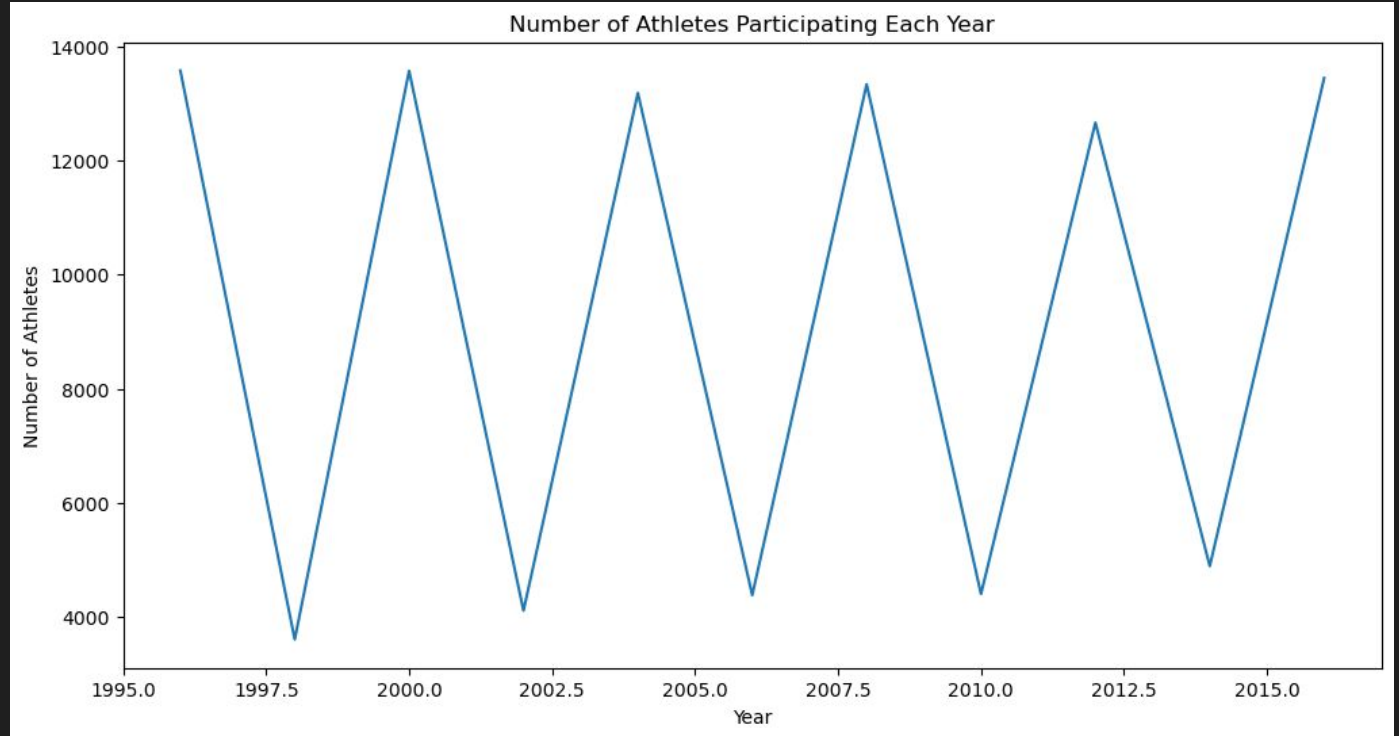
Data Visualization: Age Distribution

Age distributions are remarkably consistent across medal types. Medalists and non medalists alike cluster in their mid-twenties. The median sits around 25 for all groups, with the interquartile range roughly spanning 22-29.



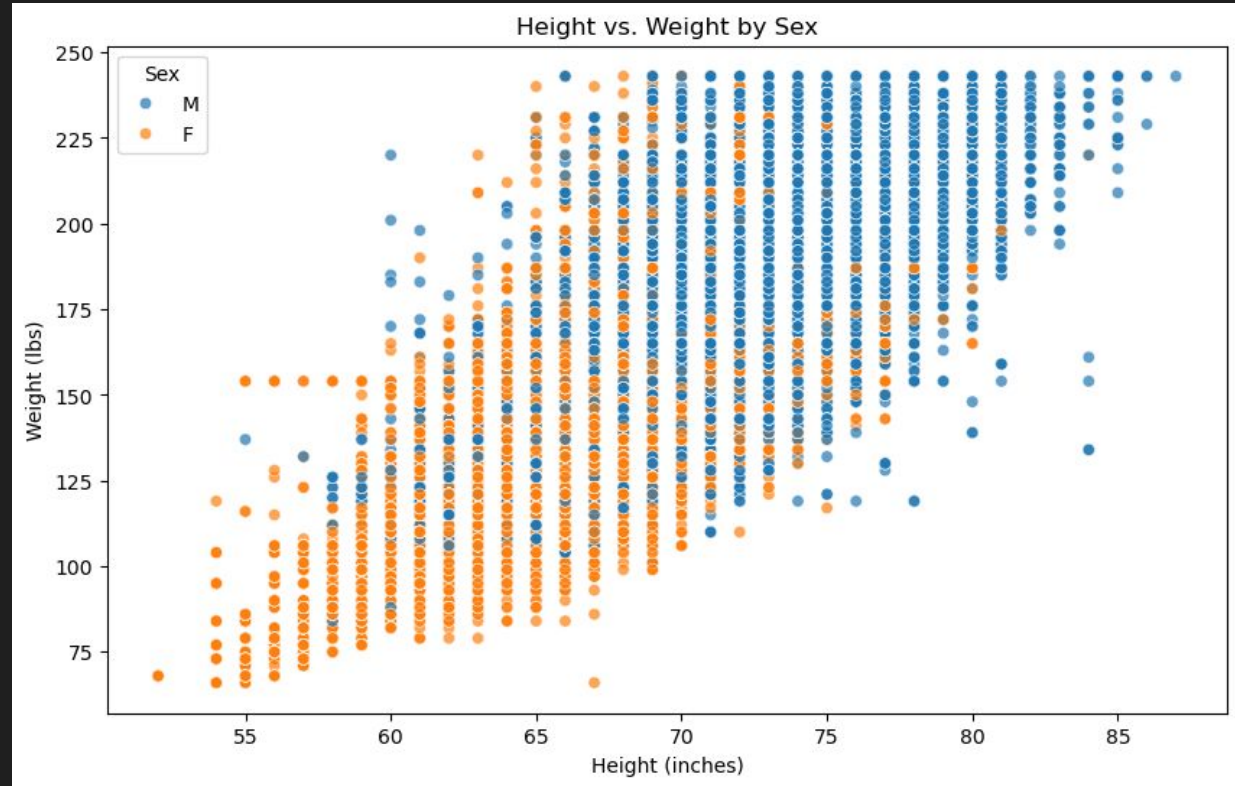
Data Visualizations: Participation Over Year

This sawtooth pattern reflects the alternating Summer/Winter game cycle. Summer games draw 13,000+ athletes while Winter Games draw 4,000 - 5,000.



Data Visualizations: Height vs. Weight by Sex

Each dot represents one athlete. The scatter shows a clear positive linear relationship between height and weight, with two distinct clusters by sex.



Conclusion



Participation Gap

Men consistently outnumber women in headcount across nearly every Olympic Games from 1996–2016,

Sports Exclusivity

Baseball and Nordic were entirely male only. While Softball, Rhythmic Gymnastics, and Synchronized Swimming were female-only,

Medal Performance

Men earned 7,719 total medals vs. women's 6,456, a gap that mirrors the overall performance imbalance. With fewer spots and events open to women fewer medals are available to win.